

ICT703 – Module 4 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice strings*
2. *Practice list handling*

Activity 1: Strings

Write a program that initialises a string variable called “inputString” and then does the following:

inputString = "Python rules!"

- a. Print the length of inputString
- b. Print a list of words in inputString
- c. Print inputString in uppercase
- d. Print the 10th character
- e. Print the position of the string “rule” in inputString
- f. Print the string after replacing the first “!” with a “?”

Challenge 1

Test your program with

inputString = "Hello World"

Do all the above steps work? If not, put conditional statements to make your program handle the errors.

Challenge 2

Write a program that reverses the string and print out the first string and the reverse string.

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Activity 2: Lists and Accumulators

- a. Expand the M&M program to print out the number of different M&Ms in the bowl.
- b. Write a program that creates a list of 100 random numbers and then print out the amount of odd and even numbers.
 - a. Hint: develop the pseudo-code first to help with your 2 step solution. You can also import the random module to help.
- c. Enhance the solution in b. by calculating the sum of all the odd numbers

Activity 3: Lists and Summing

Develop pseudo code and a program that does the following:

Print the total of all odd numbers in the following list

```
list1 = [2,3,4,5,6,4,8,9,9,8,34,32,33,37,78,79]
```

Activity 4: Counting Vowels

Develop pseudo code and a program that does the following:

Count the vowels in a given string and then print out how many of each vowel (a, e, i, o, u)